MODULE 10A · EPISODE 1

Space Exploration: From Sputnik to Apollo and Beyond

Exploration and Flight History

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Space Exploration: From Sputnik to Apollo and Beyond

This episode chronicles the dawn of the space age, from the first beep of Sputnik 1 in 1957 to the Voyager probes now exploring interstellar space. The panel traces the space race between the Soviet Union and the United States: the first satellites, the first humans in space, the tragedies and near misses along the way, and President Kennedy's challenge that culminated in Apollo 11's landing on the Moon. It then steps back to the aviation pioneers whose courage made spaceflight imaginable, and closes with the robotic probes that have served as our eyes and ears across the Solar System.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial ecologist

Learning Objectives

After this episode students should be able to:

- Explain why the launch of Sputnik 1 in 1957 began the space race, and describe the US response.
- Describe what Telstar 1 achieved and how radiation in the Van Allen Belts cut its life short.
- Distinguish orbital from suborbital flight, using Gagarin, Shepard, and Glenn as examples.
- Identify the first Soviet and American astronauts and the risks and scares of their missions.
- Explain the significance of the Apollo 1 fire and of Kennedy's 1962 Rice University speech.
- Describe the Apollo 11 landing, the roles of its three astronauts, and the problems they overcame.
- Identify the achievements of the Wright Brothers, Bessie Coleman, Charles Lindbergh, and Amelia Earhart.
- Summarize what the Pioneer, Viking, and Voyager probes revealed about the Solar System.
- Place the major events of early flight and spaceflight on a timeline.

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QUICK REFERENCE

Key Terms

Term	Definition
Space age	The era of human activity in space, which began with the launch of Sputnik 1 in 1957.
Space race	The competition between the Soviet Union and the United States to reach milestones in spaceflight first.
Artificial satellite	A human-made object placed in orbit around Earth or another body.
Sputnik 1	The first artificial satellite: a 23-inch metal sphere launched by the Soviet Union on October 4, 1957.
Vanguard TV-3	The first US attempt to launch a satellite; it failed at launch in December 1957.
Explorer 1	The first successful US satellite, launched January 31, 1958.
Telstar 1	The first transatlantic telecommunication satellite (1962); it relayed the first television images across the Atlantic.
Van Allen Belts	Zones of energetic charged particles trapped by Earth’s magnetic field; first detected by Explorer 1.
Orbital flight	A flight fast enough to circle Earth continuously, falling around the planet rather than back to it.
Suborbital flight	A flight that reaches space but lacks the speed to circle Earth, so it arcs back down.
Reentry	The return of a spacecraft through Earth’s atmosphere.
Heat shield	The protective layer on a capsule that absorbs and carries away the intense heat of reentry.
Vostok 1	Yuri Gagarin’s capsule, which made the first human orbit of Earth in 1961.
Liberty Bell 7	Gus Grissom’s capsule; it sank after its hatch blew prematurely.
Friendship 7	John Glenn’s capsule, which made the first American orbits of Earth in 1962.
Apollo program	NASA’s program to land astronauts on the Moon; six missions landed between 1969 and 1972.
Lunar module	The craft that carried two astronauts from lunar orbit to the surface and back; Apollo 11’s was the Eagle.
Command module	The crew capsule that stayed in lunar orbit and returned the crew to Earth; Apollo 11’s was Columbia.
Heavier-than-air flight	Flight by a craft denser than air, held up by wings and engine power rather than by buoyancy as a balloon is.
Barnstormer	A stunt pilot who toured the country giving air shows and rides, especially in the 1920s.
Spirit of St. Louis	Charles Lindbergh’s single-engine airplane, flown solo from New York to Paris in 1927.
Unmanned (robotic) probe	An uncrewed spacecraft sent to explore other worlds and radio its findings back to Earth.

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Term	Definition
Interstellar space	The space between the stars, beyond the bubble of solar wind that surrounds the Sun.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
October 4, 1957	Launch of Sputnik 1; the space age begins
23 inches (about 58 cm)	Diameter of the Sputnik 1 sphere
18,000 mph (about 29,000 km/h)	Sputnik 1's speed in orbit
January 31, 1958	Explorer 1, the first successful US satellite
1962	Telstar 1 launched, about four years after Explorer 1
4 months	Telstar 1's working life before radiation damage stopped it
Early 1963	Telstar 1 goes silent for good
April 12, 1961	Yuri Gagarin becomes the first human to orbit Earth
50%	Estimated chance of success for Gagarin's mission
May 5, 1961	Alan Shepard's suborbital flight, the first American in space
February 20, 1962	John Glenn becomes the first American to orbit Earth
3	Orbits completed by Glenn in Friendship 7
1962	Kennedy's "We choose to go to the Moon" speech at Rice University
1967	Apollo 1 fire kills Grissom, White, and Chaffee
77	Glenn's age when he returned to space aboard the Space Shuttle
July 20, 1969	Apollo 11's lunar module, the Eagle, lands on the Moon
Less than 1 minute	Fuel to spare when the Eagle touched down
12 men, 6 missions	Moonwalkers across the Apollo landings
1903	The Wright Brothers' first powered, controlled flight at Kitty Hawk
1927	Lindbergh's solo flight from New York to Paris
33 hours	Approximate length of Lindbergh's flight
1937	Amelia Earhart disappears attempting to fly around the globe
1976	Viking missions send the first clear images from the surface of Mars
1977	Launch of the twin Voyager spacecraft
4	Giant planets visited by Voyager 2: Jupiter, Saturn, Uranus, and Neptune
Nearly 50 years	How long both Voyagers have been operating

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EPISODE CONTENT

Part-by-Part Summary

Part 1 — Sputnik and the Start of the Space Race

The space age began on October 4, 1957, when the world awoke to the news that the Soviet Union had launched **Sputnik 1**, the first artificial satellite. It was a 23-inch (about 58 cm) metal sphere orbiting Earth at about **18,000 miles per hour**, and its simple radio beep signaled that humanity had crossed a new frontier. The **space race** was officially on.

The US response was hurried. Its first attempt, **Vanguard TV-3**, failed. Then, on **January 31, 1958**, **Explorer 1** successfully reached orbit, becoming the first US satellite.

Part 2 — Telstar 1 and the Van Allen Belts

About four years later came the real leap for the US: **Telstar 1**, the first transatlantic telecommunication satellite. Launched in 1962, it relayed the **first television images across the Atlantic**, including a press conference by President Kennedy.

Its life was cut short. Telstar 1 suffered radiation damage while passing through the **Van Allen Belts**, zones of charged particles held by Earth's magnetic field, which had been intensified by a high-altitude US nuclear test. It failed just **four months** after launch and, after a brief revival, went silent for good in early 1963.

Part 3 — The First Humans in Space

The race then shifted to putting a human in orbit, and the Soviets once again took the lead. On **April 12, 1961**, **Yuri Gagarin** became the first human to orbit the Earth. The mission was incredibly risky, with only about a **50% chance of success**. His **Vostok 1** capsule tumbled violently on reentry, but he ejected as planned and parachuted safely to the ground.

The US followed less than a month later with **Alan Shepard** on **May 5, 1961**. His was a **suborbital** flight: it reached space but did not have the speed to circle Earth, so it arced back down. It was still a crucial step, and Shepard would later walk on the Moon as commander of **Apollo 14**.

Gus Grissom was the second American in space. His flight was successful, but his capsule, the **Liberty Bell 7**, sank after its hatch blew prematurely. Grissom's story has a tragic end: in 1967 he, **Ed White**, and **Roger Chaffee** died in a fire during a launch pad test for **Apollo 1**. The devastating loss led to major safety improvements in the Apollo program.

> *† The episode says the hatch "blew prematurely." It did, but the cause was never determined. Grissom always said he did not trigger it, and investigators did not fault him.*

Part 4 — Kennedy's Challenge and John Glenn

The driving force behind the American effort was **President Kennedy**. His **1962 speech at Rice University** is legendary: "We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard." That speech galvanized the nation and NASA.

Earlier that same year, **John Glenn** had already given the nation a boost. On **February 20, 1962**, he became the **first American to orbit the Earth**, completing **three orbits** in his

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Friendship 7 capsule. Like Gagarin, he had a scare: a faulty signal suggested his **heat shield** was loose. He made it back safely, went on to a distinguished career as a US Senator, and even returned to space at **age 77** aboard the Space Shuttle.

† Kennedy first set the Moon goal in May 1961, in an address to Congress. The 1962 Rice speech explained the goal to the public and rallied support for it.

Part 5 — Apollo 11 and the Moon Landings

All of this culminated in the summer of 1969. On **July 20, 1969**, the world watched as Apollo 11's **lunar module**, the **Eagle**, touched down on the Moon. **Neil Armstrong** and **Buzz Aldrin** became the first humans to walk on another world, while **Michael Collins** orbited above in the **command module Columbia**. It fulfilled Kennedy's challenge, a monumental achievement for humanity.

The landing was fraught with tension. Armstrong had to take **manual control** to avoid a boulder field, landing with **less than a minute of fuel** to spare. On departure, a critical circuit breaker broke, and the crew had to improvise with a pen to get the engine to fire.

In the end, **12 men walked on the Moon across six Apollo missions**. It remains one of the greatest accomplishments in human history.

Part 6 — Pioneers of Flight

Spaceflight stands on the shoulders of early aviation pioneers. The **Wright Brothers** achieved the first **powered, heavier-than-air controlled flight** at Kitty Hawk in **1903**.

Bessie Coleman was the first African-American and Native-American woman to earn a pilot's license. Denied the opportunity in the US, she went to France to learn to fly and became a successful **barnstormer**.

In **1927**, **Charles Lindbergh** flew the **Spirit of St. Louis** solo from New York to Paris, a grueling **33-hour** flight that made him a global hero. **Amelia Earhart** became the first woman to fly solo across the Atlantic. Her attempt to circumnavigate the globe in **1937** ended in her mysterious disappearance, a tragedy that still captures the public imagination.

† Lindbergh's flight was the first solo nonstop crossing of the Atlantic. Others had already flown it nonstop with a two-person crew (Alcock and Brown, 1919).

Part 7 — Robotic Explorers

Alongside the crewed missions, **unmanned probes** have been our robotic eyes and ears in the cosmos. **Pioneer 10 and 11** gave us our first close-up images of **Jupiter and Saturn**.

The **Viking** missions in **1976** provided the first clear images from the surface of **Mars** and conducted experiments that some scientists thought hinted at living microbes, though most now credit ordinary soil chemistry.

The twin **Voyager** spacecraft, launched in **1977**, made a grand tour of the outer planets. **Voyager 2** visited **Jupiter, Saturn, Uranus, and Neptune**, and both Voyagers are still operating nearly 50 years later, exploring **interstellar space**.

From the first beep of Sputnik to the Voyager probes pushing the boundaries of our Solar System, the story of space exploration is one of courage, ingenuity, and an unyielding desire to reach for the stars.

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TRY IT YOURSELF

Race Lindbergh, then lap the planet. Lindbergh flew about 3,600 miles (about 5,800 km) from New York to Paris in about 33 hours. What was his average speed in miles per hour? Sputnik 1 orbited at about 18,000 mph. About how many times faster was it? At that speed, about how long would it take to travel once around Earth (circumference about 25,000 miles, or 40,000 km)?

(Answer: 3,600 miles ÷ 33 hours ≈ 110 mph, about 175 km/h. Sputnik was 18,000 ÷ 110 ≈ 160 times faster. Once around Earth: 25,000 miles ÷ 18,000 mph ≈ 1.4 hours, about 83 minutes. Sputnik actually took about 96 minutes per orbit, because its orbit was a larger circle, hundreds of miles above the ground.)

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STUDY QUESTIONS

Questions and Model Answers

Model answers follow each question. Accept answers in students' own words that capture the key ideas.

Red numbers mark core questions — the ones to be sure students can answer.

Part 1 — Sputnik and the Start of the Space Race

1. What happened on October 4, 1957, and why is that date considered the start of the space age?

> **Answer:** The Soviet Union launched Sputnik 1, the first artificial satellite. It showed that humanity had crossed a new frontier into space and officially began the space race.

2. Describe Sputnik 1: its shape, its size, and how fast it traveled.

> **Answer:** A metal sphere about 23 inches (about 58 cm) across, orbiting Earth at about 18,000 mph (about 29,000 km/h).

3. How did the United States respond to Sputnik? Name its failed first attempt and its first successful satellite, with the date.

> **Answer:** The response was hurried. The first attempt, Vanguard TV-3, failed. Explorer 1 launched successfully on January 31, 1958.

4. What was the space race?

> **Answer:** The competition between the Soviet Union and the United States to reach milestones in spaceflight first: the first satellite, the first human in orbit, and ultimately the first landing on the Moon.

Part 2 — Telstar 1 and the Van Allen Belts

1. What was Telstar 1, and what historic first did it achieve?

> **Answer:** Launched in 1962, it was the first transatlantic telecommunication satellite. It relayed the first television images across the Atlantic, including a press conference by President Kennedy.

2. What damaged Telstar 1, and what had made that hazard worse?

> **Answer:** Radiation in the Van Allen Belts, zones of charged particles trapped by Earth's magnetic field. The belts had been intensified by a high-altitude US nuclear test.

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3. How long did Telstar 1 work before failing, and when did it go silent for good?

> **Answer:** It failed about four months after launch. After a brief revival, it went silent for good in early 1963.

Part 3 — The First Humans in Space

1. Who was the first human to orbit the Earth? Give the date and the name of the capsule.

> **Answer:** Yuri Gagarin of the Soviet Union, on April 12, 1961, in Vostok 1.

2. Why was Gagarin's mission considered so risky, and what happened during his return to Earth?

> **Answer:** It was estimated to have only about a 50% chance of success. His capsule tumbled violently on reentry, but he ejected as planned and parachuted safely to the ground.

3. How was Alan Shepard's flight different from Gagarin's? Explain the difference between orbital and suborbital flight.

> **Answer:** Shepard's flight (May 5, 1961) was suborbital: it reached space but lacked the speed to circle Earth, so it arced back down. Gagarin's was orbital: he traveled fast enough to circle the whole planet.

4. What did Shepard go on to do later in his career?

> **Answer:** He walked on the Moon as commander of Apollo 14.

5. Who was the second American in space, and what happened to his capsule?

> **Answer:** Gus Grissom. His flight was successful, but his capsule, Liberty Bell 7, sank after its hatch blew prematurely.

6. What happened during the Apollo 1 launch pad test in 1967, and what was its lasting effect?

> **Answer:** A fire in the capsule killed astronauts Gus Grissom, Ed White, and Roger Chaffee. The loss led to major safety improvements in the Apollo program.

Part 4 — Kennedy's Challenge and John Glenn

1. What did President Kennedy say in his 1962 speech at Rice University, and what effect did the speech have?

> **Answer:** "We choose to go to the Moon in this decade and do the other things, not because they are easy, but because they are hard." The speech galvanized the nation and NASA behind the Moon effort.

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2. Who was the first American to orbit the Earth? Give the date, the name of the capsule, and the number of orbits.

> **Answer:** John Glenn, on February 20, 1962, in Friendship 7. He completed three orbits.

3. What scare did Glenn face during his mission?

> **Answer:** A faulty signal suggested that his heat shield was loose, which could have been fatal on reentry. He made it back safely.

4. What did Glenn do after his orbital flight?

> **Answer:** He had a distinguished career as a US Senator and returned to space at age 77 aboard the Space Shuttle.

Part 5 — Apollo 11 and the Moon Landings

1. When did Apollo 11 land on the Moon? Name the lunar module and the command module.

> **Answer:** July 20, 1969. The lunar module was the Eagle; the command module was Columbia.

2. Name the three Apollo 11 astronauts and describe each one’s role.

> **Answer:** Neil Armstrong and Buzz Aldrin landed and walked on the Moon. Michael Collins orbited above in the command module Columbia.

3. Describe two problems the Apollo 11 crew had to overcome.

> **Answer:** Armstrong had to take manual control to avoid a boulder field and landed with less than a minute of fuel to spare. On departure, a critical circuit breaker broke, and the crew improvised with a pen to get the engine to fire.

4. How many people have walked on the Moon, and across how many missions?

> **Answer:** Twelve men, across six Apollo missions.

5. In what way did Apollo 11 fulfill Kennedy’s challenge?

> **Answer:** Kennedy challenged the nation to go to the Moon within the decade; Apollo 11 landed in July 1969, before the decade was out.

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Part 6 — Pioneers of Flight

1. What did the Wright Brothers achieve, and where and when did they do it?

> **Answer:** The first powered, heavier-than-air controlled flight, at Kitty Hawk, North Carolina, in 1903.

2. Who was Bessie Coleman, and why did she go to France?

> **Answer:** The first African-American and Native-American woman to earn a pilot's license. She was denied the opportunity to learn to fly in the US, so she trained in France and became a successful barnstormer.

3. What did Charles Lindbergh accomplish in 1927?

> **Answer:** He flew the Spirit of St. Louis solo from New York to Paris, a grueling flight of about 33 hours that made him a global hero.

4. What "first" did Amelia Earhart achieve, and what happened in 1937?

> **Answer:** She was the first woman to fly solo across the Atlantic. In 1937 she disappeared during an attempt to fly around the globe.

5. Why does the episode say that spaceflight "stands on the shoulders" of early aviation pioneers?

> **Answer:** Accept reasoned answers: the pioneers proved that controlled, powered flight was possible, pushed its limits of distance and endurance, and showed the courage that later made spaceflight imaginable.

Part 7 — Robotic Explorers

1. What is an unmanned probe, and why does the episode call probes our "robotic eyes and ears"?

> **Answer:** An uncrewed spacecraft sent to explore other worlds. Its cameras and instruments see and measure places humans cannot yet reach, and it radios the results back to Earth.

2. What did the Pioneer 10 and 11 missions give us?

> **Answer:** Our first close-up images of Jupiter and Saturn.

3. What did the Viking missions accomplish in 1976? What did some scientists think their experiments showed, and what do most scientists think now?

> **Answer:** They sent back the first clear images from the surface of Mars. Some scientists thought their experiments hinted at living microbes, but most now credit ordinary soil chemistry.

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4. When were the Voyager spacecraft launched, which planets did Voyager 2 visit, and where are the Voyagers now?

> **Answer:** 1977. Voyager 2 visited Jupiter, Saturn, Uranus, and Neptune. Both are still operating, nearly 50 years later, exploring interstellar space.

5. According to the episode, what three qualities define the story of space exploration?

> **Answer:** Courage, ingenuity, and an unyielding desire to reach for the stars.

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FOR THE INSTRUCTOR

Corrections and Cautions

1. Grissom's hatch: the cause is unknown (about 2:43).
The episode says Liberty Bell 7 sank "after the hatch blew prematurely." That is true, but why it blew was never determined. Grissom always said he did not trigger it, and investigators did not fault him. The capsule was recovered from the Atlantic floor in 1999. Footnoted under the Part 3 summary in both guides.

2. Kennedy set the goal in 1961 (about 3:06).
Calling Kennedy the "driving force" is fair, but the Moon commitment came in his May 25, 1961 address to Congress. The September 1962 Rice speech explained the goal and rallied the public. Students sometimes assume the Rice speech launched the program. Footnoted under the Part 4 summary.

3. Lindbergh: first solo nonstop crossing (about 5:11).
Alcock and Brown flew the Atlantic nonstop in 1919 with a crew of two. Lindbergh's 1927 achievement was the first solo nonstop crossing (about 33½ hours). Footnoted under the Part 6 summary.

4. Telstar's "first" needs its qualifier (about 1:27).
Echo 1 (1960) had already bounced signals as a passive reflecting balloon. Telstar 1, built by AT&T;'s Bell Labs and launched by NASA in July 1962, was the first active satellite to relay live television across the Atlantic. "Four years later" is approximate: about four and a half years after Explorer 1.

5. The Starfish Prime connection (about 1:46).
The high-altitude US test was Starfish Prime, detonated July 9, 1962, the day before Telstar's launch. The episode's account of the radiation damage is accurate.

6. Gagarin's "50% chance" is an estimate (about 2:14).
The figure is a widely repeated estimate, not an official calculation. Treat it as an illustration of the risk, not a precise number.

7. Gagarin's ejection was kept secret (about 2:19).
"As planned" is correct: Vostok was designed for the pilot to eject and land separately. The Soviets concealed this for years, because record rules at the time expected the pilot to land with the craft.

8. The pen on Apollo 11 (about 4:29).
The plastic knob of the ascent-engine arming circuit breaker had broken off, likely bumped by a backpack. Aldrin pushed the breaker in with a felt-tip pen, chosen because it would not conduct electricity. The breaker armed the engine; it did not fire it directly.

9. "Less than a minute of fuel" (about 4:22).
Commonly cited figures range from about 25 to 45 seconds of usable descent fuel, depending on how the margin is counted. The episode's phrasing is safe.

10. Kitty Hawk is the traditional name (about 4:49).
The Wright Brothers' flights on December 17, 1903 took place at Kill Devil Hills, about four miles from the town of Kitty Hawk, North Carolina.

11. Bessie Coleman's license (about 4:58).
She earned it in France in 1921 from the international aviation federation (FAI), the first African-American woman and the first woman of Native-American descent to hold a pilot's license.

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12. Viking: “first clear images” is the right qualifier (about 5:50).
The Soviet Mars 3 lander (1971) touched down first but sent only a brief, indistinct partial image before failing. Viking’s were the first clear surface images.

13. The Voyagers today (about 6:11).
Only Voyager 2 visited all four giants; Voyager 1 flew past Jupiter and Saturn. Voyager 1 crossed into interstellar space in 2012 and Voyager 2 in 2018. Both still operate, but several instruments have been switched off to conserve dwindling power.

FOR THE INSTRUCTOR

Teaching Notes

Build a timeline.
The episode is not strictly chronological: it reaches the Moon, then steps back to 1903. Have students place about fifteen events from Numbers to Know on a single timeline from 1903 to the present.

Connect Explorer 1 to Telstar.
Explorer 1’s Geiger counter, built by James Van Allen’s team, discovered the radiation belts in 1958. Four years later those same belts, pumped up by Starfish Prime, killed Telstar 1. It is a tidy cause-and-effect arc.

Orbital versus suborbital.
Use Newton’s cannonball: fire faster and faster until the ball falls around Earth instead of back to it. Shepard reached roughly 5,000 mph; orbit needs about 17,500 mph. Glenn’s three orbits followed nine months later.

Play the Rice speech.
The “We choose to go to the Moon” passage is under a minute long and freely available from the JFK Library. Ask students why Kennedy stressed that the goal was hard.

Voyager 1 reaches one light-day.
Around November 17–18, 2026, Voyager 1 becomes the first spacecraft one light-day from Earth: a command takes a full day to arrive and the reply another full day. It is a timely hook for the closing segment.

The cost of exploration.
Gagarin’s odds, Grissom’s near-drowning, Apollo 1, and Earhart’s disappearance all appear in one episode. A short discussion of why people accept such risks works well before the robotic-probe segment.

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